Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?
______________________show ip route ospf____ 2

Part 3: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ______Serial/0/0/1______ 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _65(serial link cost64+Ethernet cost1)_________ 2

Does R2 show up as an OSPF neighbor on R1? No________ 0

Does R2 show up as an OSPF neighbor on R3? __No______ 2

What does this information tell you?

______Passive interfaces cause neighbor relationships to break down and other routers cannot learn routes through this router.____ 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

___R2(config)# router ospf 1R2(config-router)# no passive-interface s0/0/1_____________________ 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____Serial/0/0/1_______ 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

________65(serial link cost64+Ethernet cost1)____

Is R2 listed as an OSPF neighbor to R3? __Yes______

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

___The default reference bandwidth is 100 Mb/s, which is insufficient to differentiate between gigabit links. Changing the reference bandwidth allows the OSPF cost to more accurately reflect the actual speed of each link.____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

____The OSPF costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 are both calculated using the formula **cost = reference bandwidth (100 Mbps) / interface bandwidth (in Mbps)**.

- For 192.168.3.0/24 (via GigabitEthernet interface, 1000 Mbps): 100 / 1000 = 0.1 → rounded to minimum cost of **1**.
- For 192.168.23.0/30 (via the point-to-point link interface): depending on the actual interface bandwidth (commonly 100 Mbps or 1000 Mbps in labs), the cost is **1** (or higher if slower bandwidth is configured). With default reference bandwidth, high-speed links cannot be properly differentiated, which is why the reference bandwidth is often increased in modern networks.

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___The new accumulated cost to the 192.168.23.0/24 network on R1 ~~is **7813**~~ 0

This is because:

- Each serial interface now has bandwidth = 128 kbps.
- With reference bandwidth = 100 Mbps (100,000 kbps), interface cost = 100,000 / 128 ≈ 781 (rounded down).
- The total (accumulated) cost is the sum of costs on all outgoing interfaces along the best OSPF path from R1 to 192.168.23.0/24.

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Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_______The ip ospf cost 1565 command sets the cost of R1S0/0/1 to 1565, making it more costly to go directly through this interface than the path through R2. This causes OSPF to choose the lower cost, path through R2____

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   ____In OSPF, the router ID is a unique identifier that identifies each router and maintains adjacency. Conflicting Router IDs can lead to confusing routing information____
   ______

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2. Why is the DR/BDR election process not a concern in this lab?

   _____In this experiment, we mainly use point-to-point links with a simpler topology, and DR/BDR is only used for broadcast networks____
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3. Why would you want to set an OSPF interface to passive?

   ___

   ___This is because it allows the interface to stop sending OSPF hello packets and to stop establishing adjacencies, which reduces unnecessary routing update traffic and avoids leaking routing information____ _______

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   ____OSPF uses zones to divide the network, splitting the entire network into smaller parts through zone 0 (the backbone zone), a layered design that improves the scalability of the network.____ ___

   area: terminology

   3

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   ____router ospf 1network 10.0.0.0 0.255.255.255 area 0_______________________________

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   ____Since EIGRP has an AD of 90, OSPF has an AD of 110, and RIP has an AD of 120, the EIGRP route with the lowest AD is selected.____

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